

Conditional Probability and Base Rates

& an introduction to expected value

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Quantitative Reasoning II · INF 1220 · University of Austin

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Probability Rules

$$\sum_i P(A_i) = 1$$

all possible outcomes sum to 1

$$P(\text{not } A) = 1 - P(A)$$

$$P(A \text{ or } B) = P(A) + P(B)$$

only if A and B are mutually exclusive

$$P(A \text{ and } B) = P(A) \times P(B)$$

only if A and B are independent

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

a count over a *smaller* total

$$P(A \mid B) \neq P(B \mid A)$$

We calculate probabilities by counting the number of favorable outcomes and dividing by the total number of possible outcomes.

Some Claims

1. "This test is 98.5% accurate. It came back positive."
2. "Most successful founders dropped out of college."

Does the Stated Probability Give You the Answer You Need?

What was said	What you wanted to know
$P(\text{dropped out} \mid \text{successful})$	$P(\text{successful} \mid \text{dropped out})$
$P(\text{positive} \mid \text{has it})$	$P(\text{has it} \mid \text{positive})$

Routine Screening

The U.S. Preventive Services Task Force recommends a mammogram every two years for women aged 40–74, including women with no symptoms.

A woman in that group gets a call back form the lab with a positive result.

What is the probability she has cancer?

Real Conditional Probabilities

A woman gets a call back from the lab with a positive result from her routine mammogram. What is the probability she has cancer? *Note: We're using approximate rates to make the math easier.*

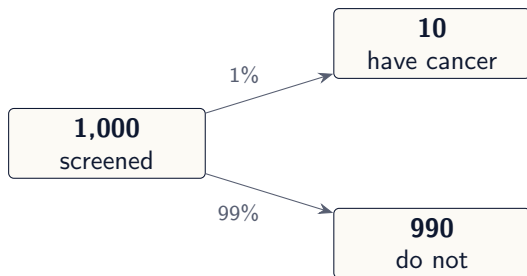
Sensitivity	90%	of cancers are caught
Specificity	88.9%	of healthy women are correctly cleared
Base rate	10 cancers per 1,000	women screened

Real BCSC figures (2007–2013): 86.9% sensitivity, 88.9% specificity, about 6 cancers per 1,000, PPV 4.4%.

Real Conditional Probabilities

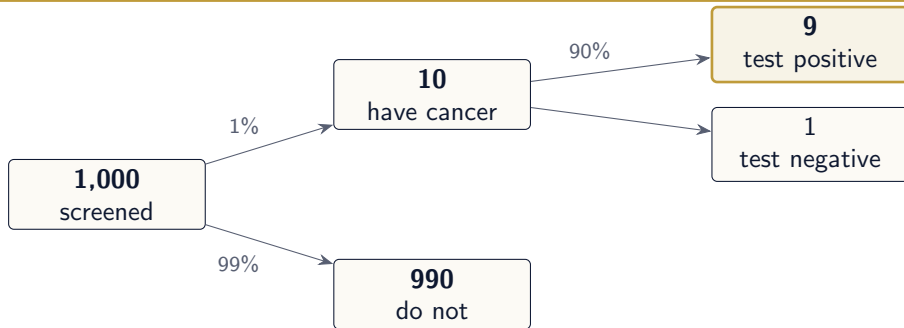
1,000
screened

Real Conditional Probabilities



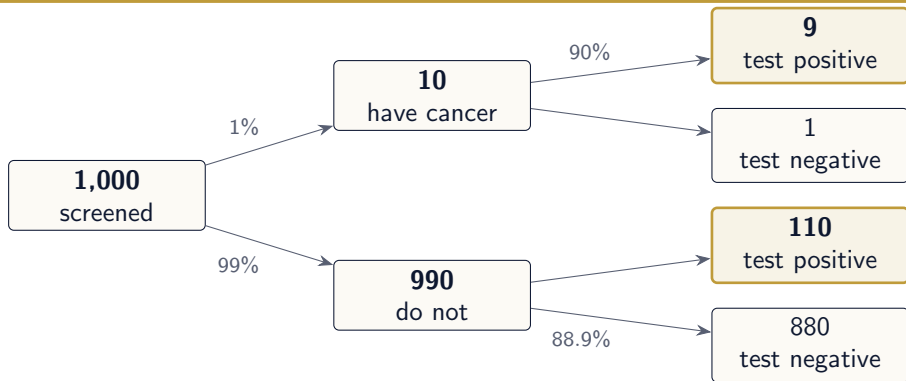
$1,000 \times 0.01 = 10$ have cancer; 990 do not

Real Conditional Probabilities



$1,000 \times 0.01 = 10$ have cancer; 990 do not
 $10 \times 0.90 = 9$ test positive

Real Conditional Probabilities

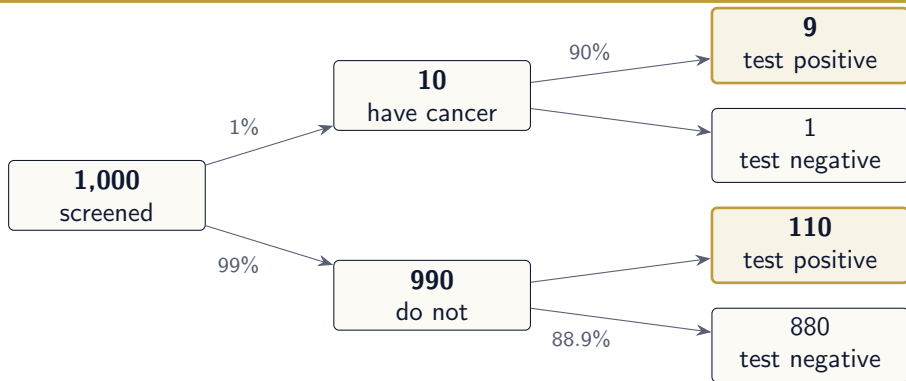


$1,000 \times 0.01 = 10$ have cancer; 990 do not

$10 \times 0.90 = 9$ test positive

$990 \times 0.889 = 880$ test negative; $990 - 880 = 110$ test positive

Real Conditional Probabilities



$1,000 \times 0.01 = 10$ have cancer; 990 do not

$10 \times 0.90 = 9$ test positive

$990 \times 0.889 = 880$ test negative; $990 - 880 = 110$ test positive

119 women are called by the lab. Only 9 of them actually have cancer.

What Is the Probability That the Woman Has Cancer?

$$P(\text{cancer} \mid \text{positive}) = \frac{9}{119} = 0.076$$

About 1 in 13.

Two Different Questions

What the test reports

$$P(\text{positive} \mid \text{cancer}) = 0.90$$

Of women with cancer, most test positive.

What the patient wants

$$P(\text{cancer} \mid \text{positive}) = \frac{9}{119} = 0.076$$

Of women who test positive, very few have cancer.

Problem 1

Workplace Drug Test

5% of employees use the drug. The test is 95% sensitive and 95% specific.

An employee tests positive. What is the probability they use the drug?

Start with 10,000 employees.

Problem 1

$$\text{use the drug} = 10,000 \times 0.05 = 500$$

$$\text{do not} = 10,000 \times 0.95 = 9,500$$

$$\text{true positives} = 500 \times 0.95 = 475$$

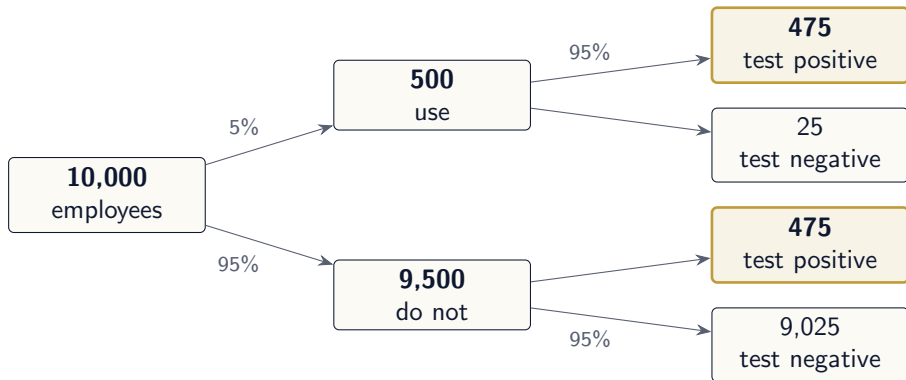
$$\text{true negatives} = 9,500 \times 0.95 = 9,025$$

$$\text{false positives} = 9,500 - 9,025 = 475$$

$$\text{positives} = 475 + 475 = 950$$

$$P(\text{uses drug} \mid \text{positive}) = \frac{475}{950} = 0.50$$

Problem 1



$$P(\text{uses drug} \mid \text{positive}) = \frac{475}{950} = 0.50$$

Problem 2

A Rare Disease

1 in 10,000 people have it. The test catches **every** case (100% sensitive) and is 99% specific.

You test positive. What is the probability you have it?

Start with 1,000,000 people.

Problem 2

$$\text{have it} = 1,000,000 \times \frac{1}{10,000} = 100$$

$$\text{do not} = 1,000,000 - 100 = 999,900$$

$$\text{true positives} = 100 \times 1.00 = 100$$

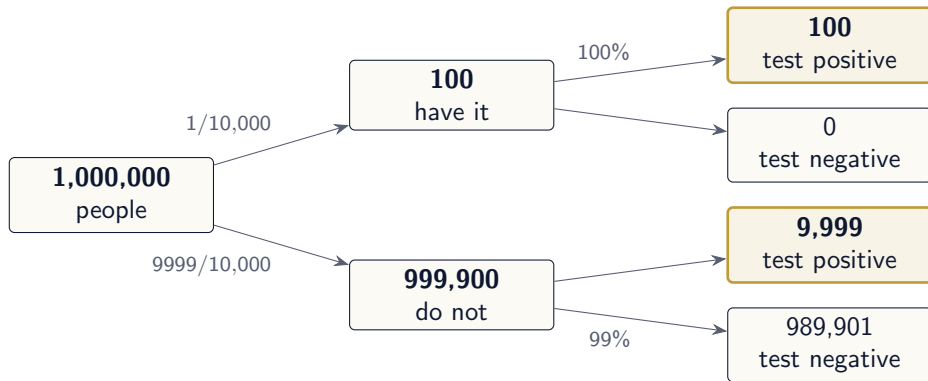
$$\text{true negatives} = 999,900 \times 0.99 = 989,901$$

$$\text{false positives} = 999,900 - 989,901 = 9,999$$

$$\text{positives} = 100 + 9,999 = 10,099$$

$$P(\text{have it} \mid \text{positive}) = \frac{100}{10,099} \approx 0.0099$$

Problem 2



$$P(\text{have it} \mid \text{positive}) = \frac{100}{10,099} \approx 0.0099$$

Prior and Posterior: Probability and Belief Revision



- The base rate is the **prior** — what you believed beforehand
- The updated number is the **posterior**
- Evidence moves the prior. It does not replace it

Expected Value

Expected Value

A probability tells you how likely something is, but doesn't tell you about the value of the outcome.

Expected value

Expected value is a weighted average of the possible outcomes.

$$E[X] = \sum_i x_i P(x_i)$$

The average result if the situation were repeated many times.

Example: A Thanksgiving Flight

A Thanksgiving flight is \$280 if you buy this week. If you wait, there is an 80% chance the fare is \$400 and a 20% chance it drops to \$220.

Outcome	Probability	Cost	Product
Fare goes up	0.80	-\$400	-\$320
Fare goes down	0.20	-\$220	-\$44
Expected cost if you wait			-\$364
Cost if you buy this week			-\$280

Problem 3

A Game

It costs \$5 to play.

- 1% chance of winning \$100
- 10% chance of winning \$10
- Otherwise nothing

What is the expected value of playing?

Problem 3

$$E[\text{winnings}] = 0.01 \times \$100 + 0.10 \times \$10 + 0.89 \times \$0 = \$2$$

$$E[\text{playing}] = \$2 - \$5 = -\$3$$

Problem 4

Routine Screening, as a Decision

Of every 1,000 women screened, 9 cancers are found, 1 is missed, 110 healthy women are called back, and 880 are correctly cleared.

Suppose a mammogram costs \$150 and follow-up imaging costs an additional \$500.

What is the expected cost per woman screened?

Problem 4

Outcome	Probability	Cost	Product
Cancer, tests positive	0.009	\$650	\$5.85
Cancer, tests negative	0.001	\$150	\$0.15
Healthy, tests positive	0.110	\$650	\$71.50
Healthy, tests negative	0.880	\$150	\$132.00
Expected cost per woman screened			\$209.50

Summary

- The probability that gets stated is often not the one you need
- $P(A | B) \neq P(B | A)$
- $P(\text{cancer} | \text{positive}) = \text{true positives} / \text{all positives}$
- The base rate is our best guess before we see the test result. A test result updates our belief.
- Expected value: weight each outcome by its probability
- The costs you leave out of an expected value decide the answer